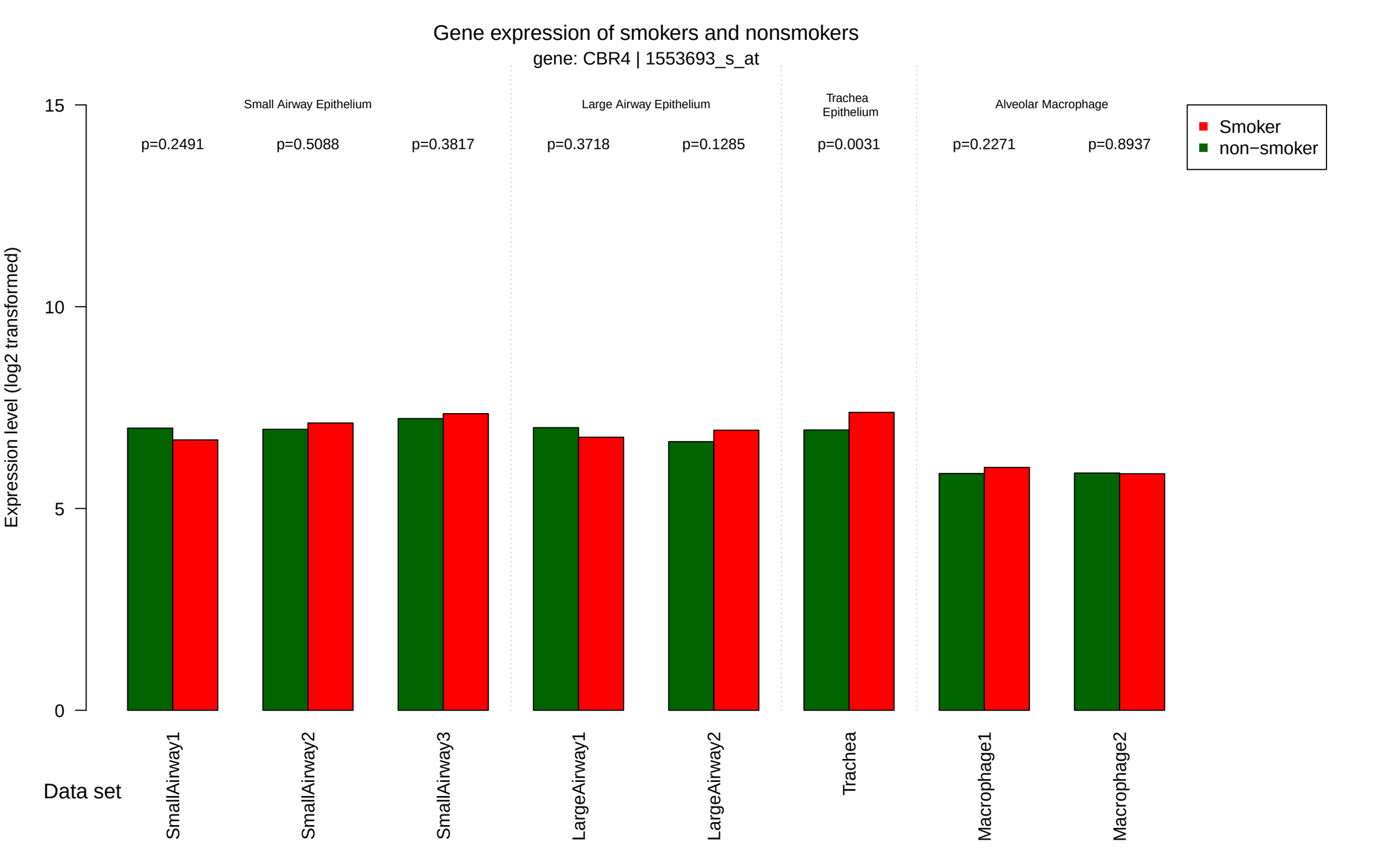
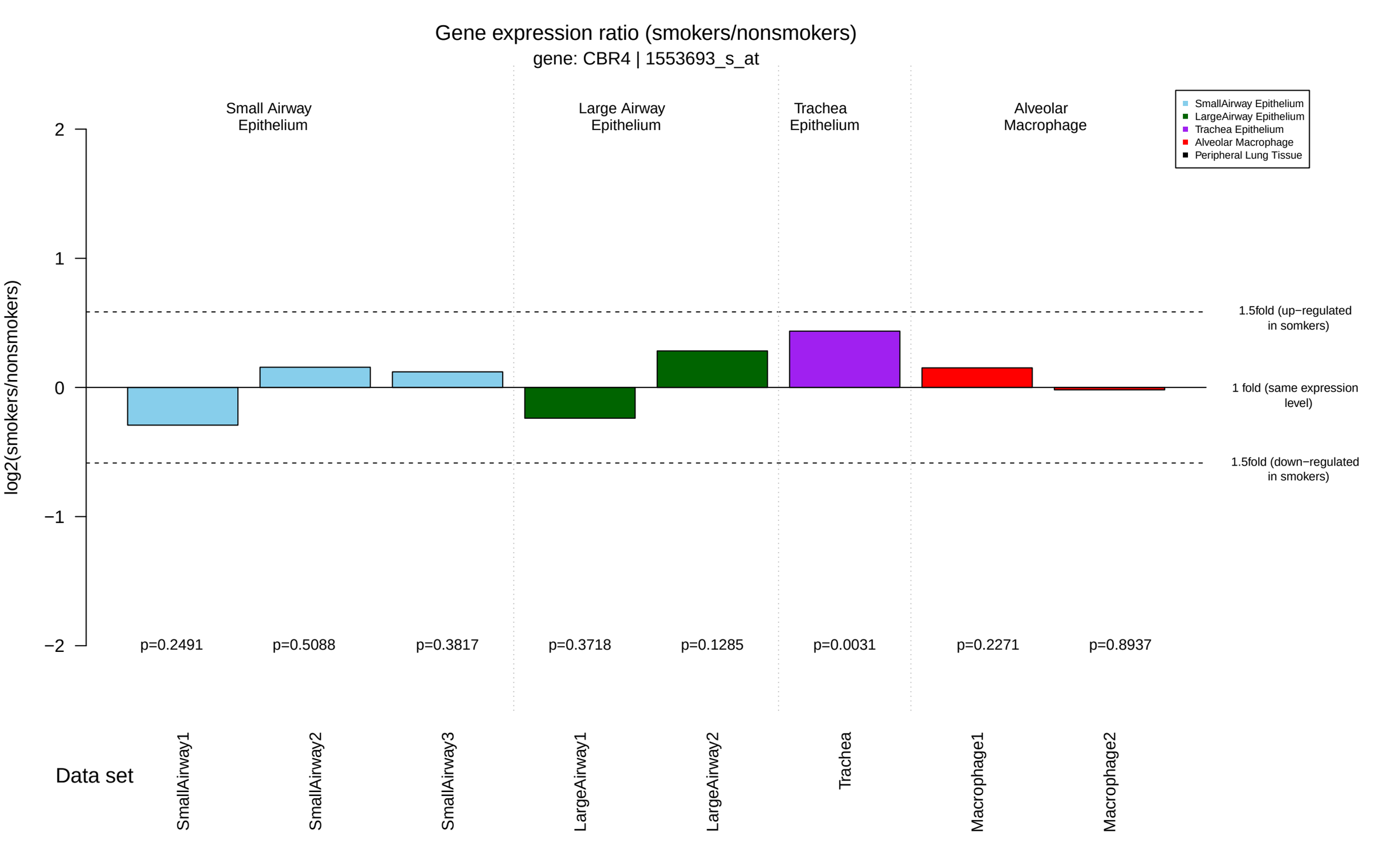
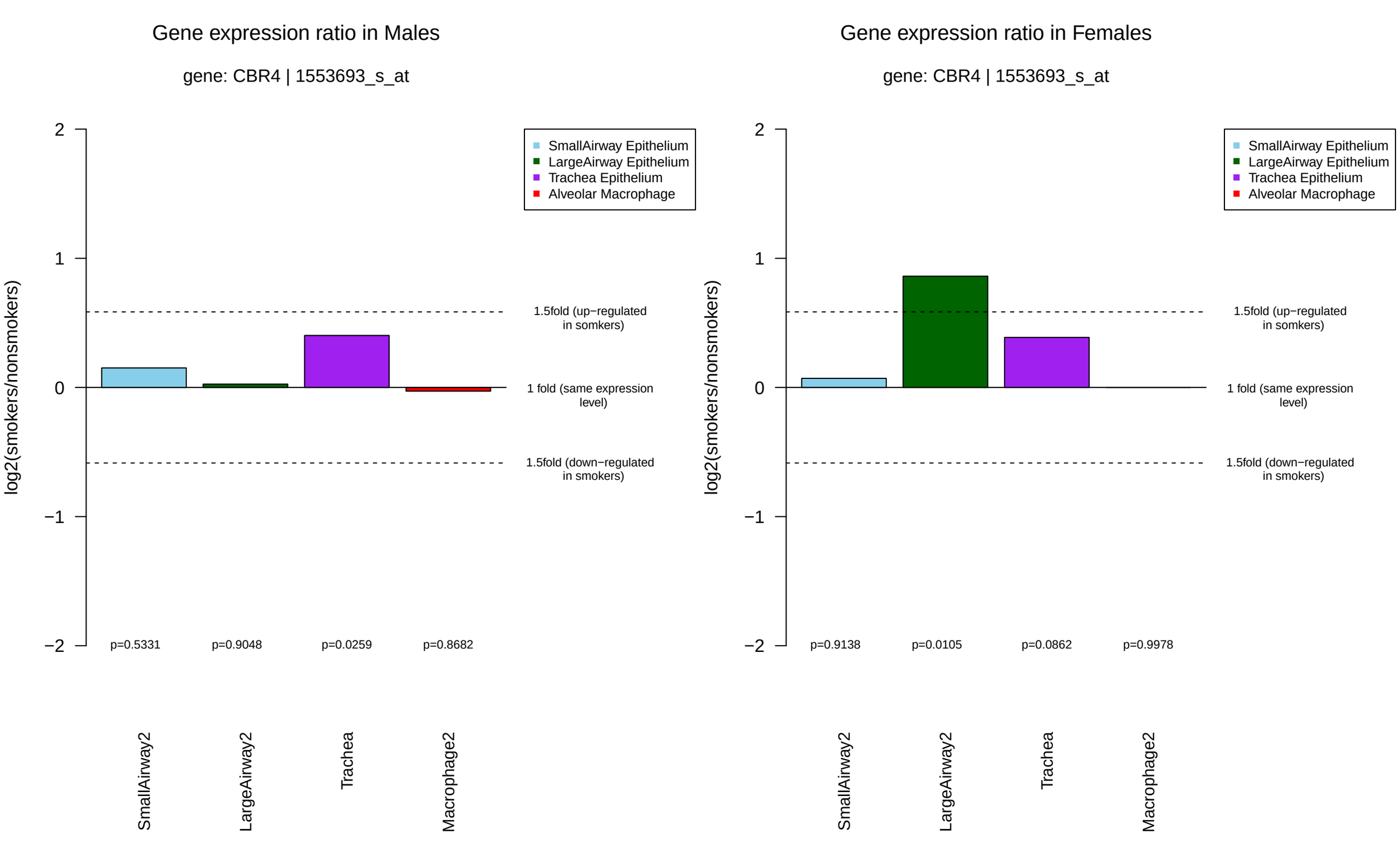
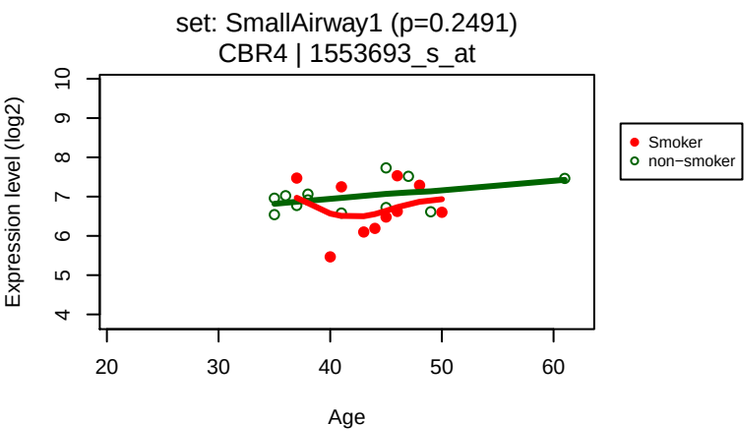
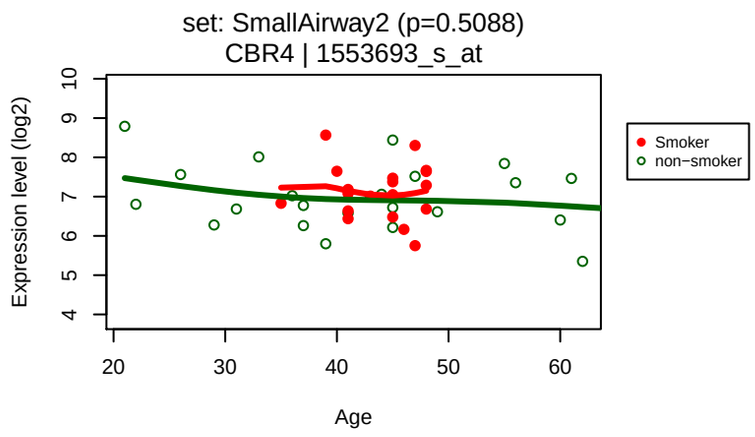
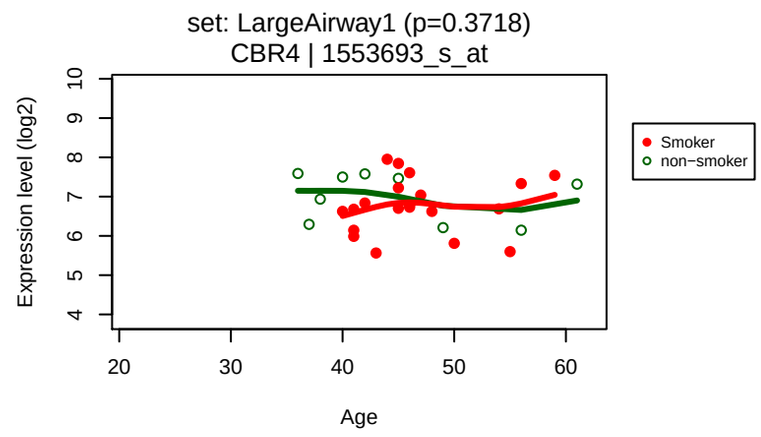

Gene expression of smokers and nonsmokers
gene: CBR4 | 1553693_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2491
p=0.5088
p=0.3817
p=0.3718
p=0.1285
p=0.0031
p=0.2271
p=0.8937
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

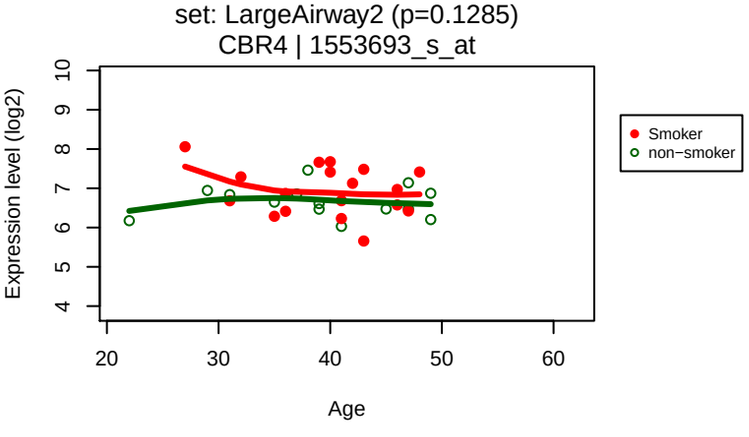

Gene expression ratio (smokers/nonsmokers)
gene: CBR4 | 1553693_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2491
p=0.5088
p=0.3817
p=0.3718
p=0.1285
p=0.0031
p=0.2271
p=0.8937
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

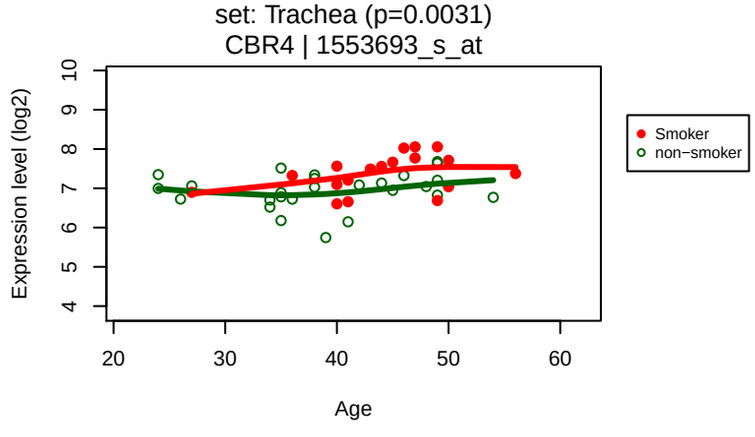

Gene expression ratio in Males
gene: CBR4 | 1553693_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5331
p=0.9048
p=0.0259
p=0.8682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CBR4 | 1553693_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9138
p=0.0105
p=0.0862
p=0.9978
SmallAirway2
LargeAirway2
Trachea
Macrophage2

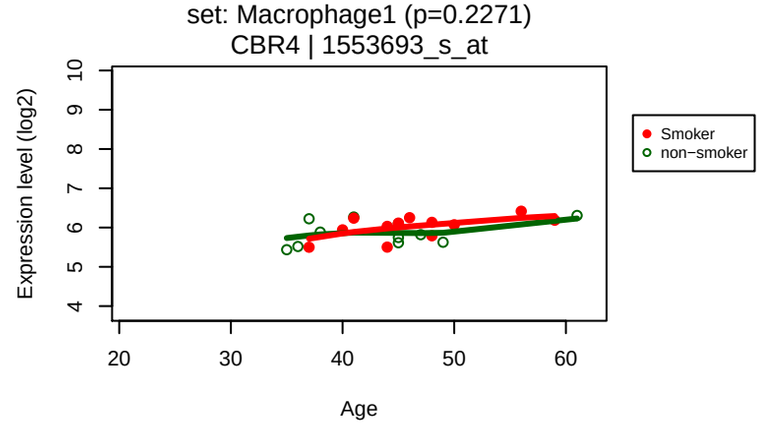

set: SmallAirway1 (p=0.2491)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker

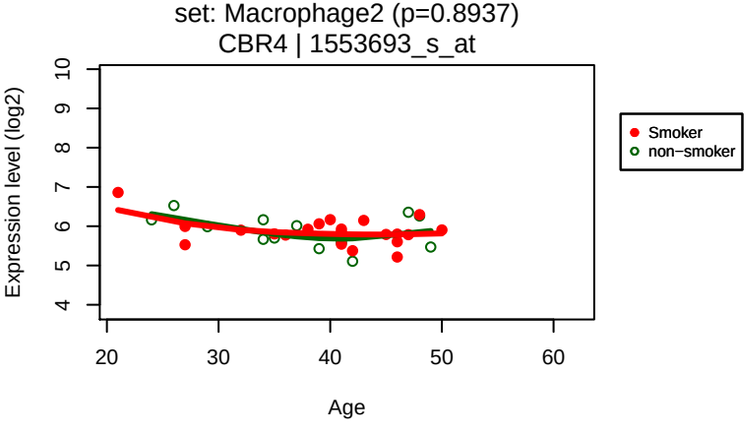

set: SmallAirway2 (p=0.5088)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker

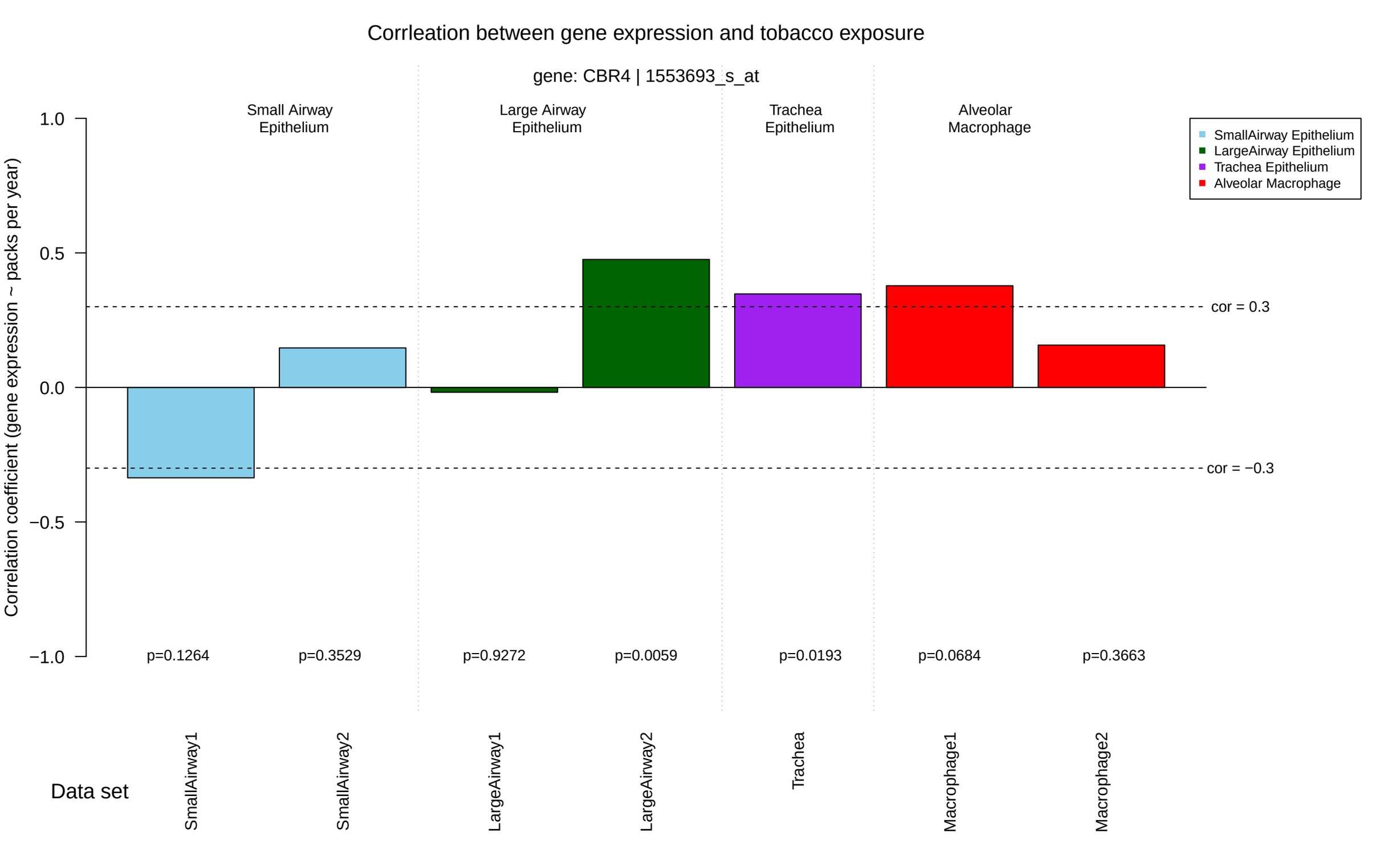

set: LargeAirway1 (p=0.3718)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1285)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0031)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2271)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8937)
CBR4 | 1553693_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CBR4 | 1553693_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1264
p=0.3529
p=0.9272
p=0.0059
p=0.0193
p=0.0684
p=0.3663
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2